

# NETWORK PACKET SNIFFER WITH ANOMALY DETECTION AND ALERT SYSTEM

## Abstract

With the rapid growth of internet usage and digital communication, networks have become increasingly vulnerable to cyber threats such as port scanning, denial-of-service attacks, and unauthorized access attempts. Monitoring network traffic in real time is therefore essential for identifying suspicious activities and ensuring network security. This project presents the development of a **Real-Time Network Packet Sniffer with an Anomaly Detection and Alert System** using Python.

The system captures network packets using packet sniffing techniques and extracts key packet attributes such as source IP address, destination IP address, protocol type, and packet length. The captured data is stored in a SQLite database for logging and further analysis. An anomaly detection mechanism is implemented to identify abnormal traffic patterns such as excessive packet transmission from a particular source or unusual communication behavior. When suspicious activity is detected, the system generates alerts to notify the user. Additionally, network traffic patterns are visualized using graphical plots for easier analysis. This project demonstrates how packet sniffing and traffic analysis can be used as an effective method for network monitoring and threat detection.

## Introduction

Network security has become a critical concern due to the increasing number of cyber attacks targeting computer systems and communication networks. Attackers often exploit network vulnerabilities through activities such as port scanning, brute force attacks, or traffic flooding to gain unauthorized access or disrupt services. To defend against such threats, organizations rely on network monitoring tools that analyse traffic patterns and detect malicious behavior.

Packet sniffing is a widely used technique in network security that involves capturing and analysing packets transmitted across a network. Security analysts use packet sniffers to examine network communication and identify abnormal activities that may indicate potential attacks.

The objective of this project is to design and implement a **network packet sniffer capable of monitoring real-time traffic and detecting anomalies**. The developed system captures packets using the Scapy library in Python and records important packet attributes for analysis. By applying simple anomaly detection techniques, the system identifies suspicious patterns and generates alerts to inform the user. This project provides practical exposure to network traffic analysis and demonstrates how cybersecurity professionals monitor network activity to detect potential threats.

## Tools Used

| Tool               | Purpose                                              |
|--------------------|------------------------------------------------------|
| Python             | Main programming language used to develop the system |
| Scapy              | Captures and analyzes network packets                |
| SQLite             | Stores packet logs and traffic data                  |
| Matplotlib         | Visualizes network traffic statistics                |
| Visual Studio Code | Development environment used for coding and testing  |

## Steps Involved in Building the Project

- Environment Setup**  
Python and Visual Studio Code were used to set up the development environment. Required libraries such as Scapy and Matplotlib were installed.
- Packet Capture**  
The Scapy library was used to capture network packets in real time from the network interface.
- Packet Information Extraction**  
Important packet details such as source IP, destination IP, protocol type, and packet length were extracted from captured packets.
- Database Logging**  
Packet information was stored in a SQLite database to maintain logs of network activity for analysis.
- Anomaly Detection**  
The system analyzes traffic patterns to detect suspicious behavior such as excessive packets from a single source or possible port scanning attempts.
- Alert Generation**  
When abnormal traffic exceeds a defined threshold, the system generates alerts to notify the user about potential threats.
- Traffic Visualization**  
Network traffic data is visualized using graphs to help understand traffic patterns and identify unusual activity.

## Conclusion

This project demonstrates the development of a **basic network monitoring system** capable of capturing packets, analyzing traffic patterns, and detecting suspicious activities. By logging packet data and generating alerts, the system helps identify potential network threats. The project highlights the importance of network traffic analysis in cybersecurity and provides practical experience in packet sniffing and anomaly detection. Such systems can assist security analysts in monitoring networks and improving overall security.